

PRAKTIKUM BASIS DATA

MODUL 08

JURNAL



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL

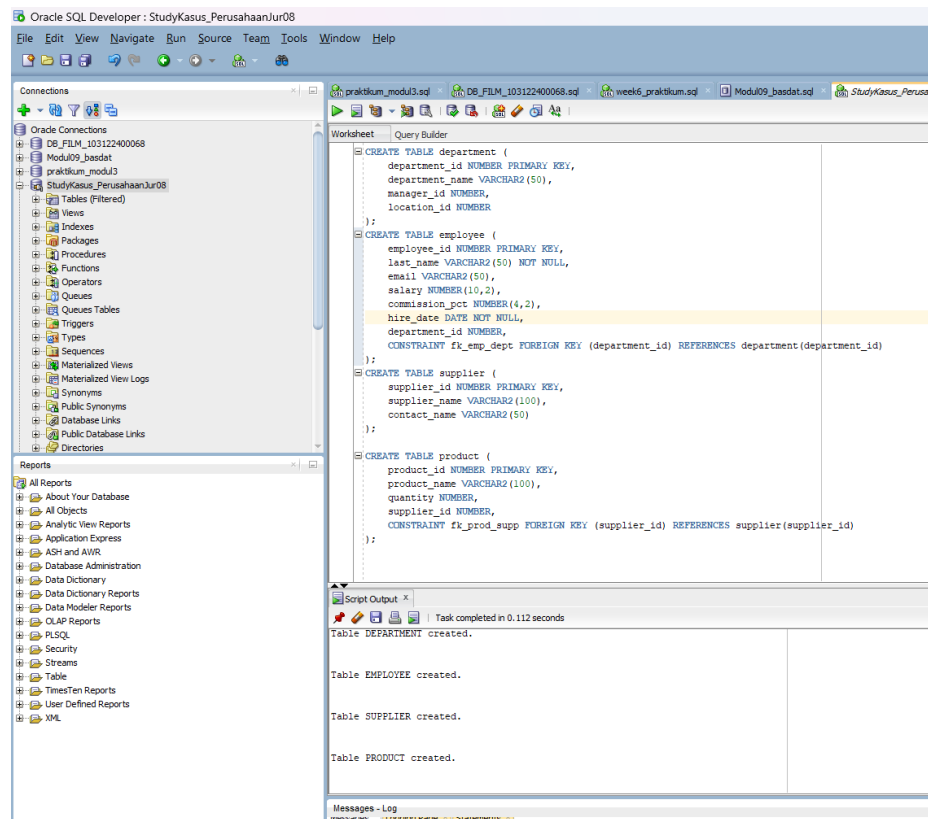
1. Soal

8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
 - a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
 - b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
 - c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Jawaban:

1.



2.

```
ANALYZE TABLE department COMPUTE STATISTICS;  
ANALYZE TABLE employee COMPUTE STATISTICS;  
ANALYZE TABLE supplier COMPUTE STATISTICS;  
ANALYZE TABLE product COMPUTE STATISTICS;
```

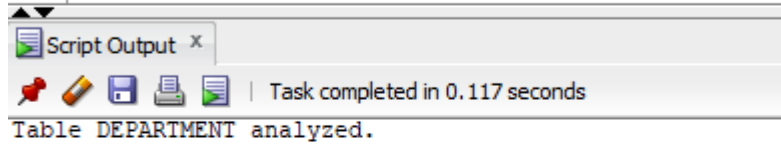


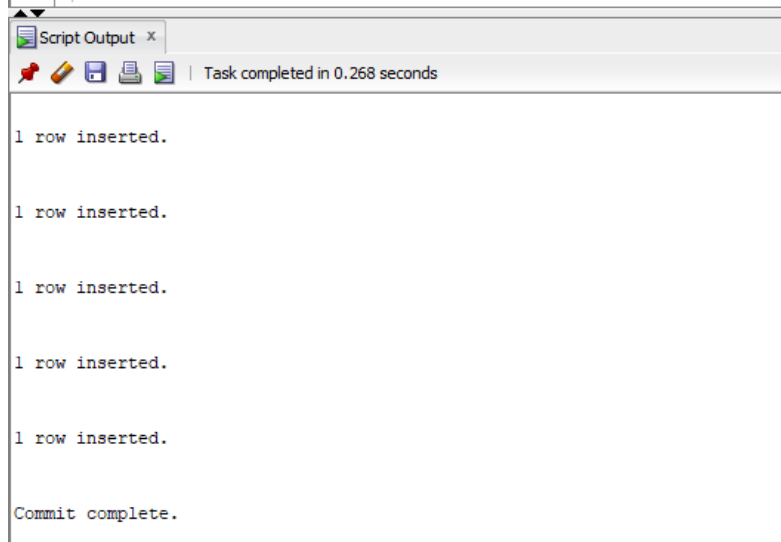
Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.

Table PRODUCT analyzed.

3.

```
INSERT INTO department VALUES (10, 'Administration', 200, 1700);  
INSERT INTO department VALUES (20, 'Marketing', 201, 1800);  
INSERT INTO department VALUES (30, 'Purchasing', 114, 1700);  
INSERT INTO department VALUES (40, 'Human Resources', 203, 2400);  
INSERT INTO department VALUES (50, 'Shipping', 121, 1500);  
INSERT INTO department VALUES (60, 'IT', 103, 1400);  
INSERT INTO department VALUES (70, 'Public Relations', 204, 2700);  
INSERT INTO department VALUES (80, 'Sales', 145, 2500);  
INSERT INTO department VALUES (90, 'Executive', 100, 1700);  
INSERT INTO department VALUES (100, 'Finance', 108, 1700);  
COMMIT;
```



```
INSERT INTO product VALUES (1001, 'Laptop Asus', 50, 3);
INSERT INTO product VALUES (1002, 'Mouse Logitech', 150, 3);
INSERT INTO product VALUES (1003, 'Kertas A4', 500, 1);
INSERT INTO product VALUES (1004, 'Tinta Printer', 200, 2);
INSERT INTO product VALUES (1005, 'Meja Kantor', 20, 5);
INSERT INTO product VALUES (1006, 'Kursi Ergonomis', 25, 5);
INSERT INTO product VALUES (1007, 'Flashdisk 64GB', 100, 4);
INSERT INTO product VALUES (1008, 'Proyektor EPSON', 10, 3);
INSERT INTO product VALUES (1009, 'Papan Tulis', 15, 6);
INSERT INTO product VALUES (1010, 'Spidol Boardmarker', 300, 1);
COMMIT;
```

Script Output x

Task completed in 0.107 seconds

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

Commit complete.

```
INSERT INTO supplier VALUES (1, 'PT Maju Bersama', 'Budi');
INSERT INTO supplier VALUES (2, 'CV Abadi Jaya', 'Andi');
INSERT INTO supplier VALUES (3, 'Global Tech', 'Sarah');
INSERT INTO supplier VALUES (4, 'Indo Supply', 'Doni');
INSERT INTO supplier VALUES (5, 'Sumber Rejeki', 'Siti');
INSERT INTO supplier VALUES (6, 'Mega Karya', 'Bambang');
INSERT INTO supplier VALUES (7, 'Lintas Nusa', 'Rina');
INSERT INTO supplier VALUES (8, 'Prima Sentosa', 'Agus');
INSERT INTO supplier VALUES (9, 'Bintang Timur', 'Lina');
INSERT INTO supplier VALUES (10, 'Karya Mandiri', 'Joko');
COMMIT;
```

Script Output x

Task completed in 0.121 seconds

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

Commit complete.

4.

```
ANALYZE TABLE department COMPUTE STATISTICS;  
ANALYZE TABLE employee COMPUTE STATISTICS;  
ANALYZE TABLE supplier COMPUTE STATISTICS;  
ANALYZE TABLE product COMPUTE STATISTICS;
```

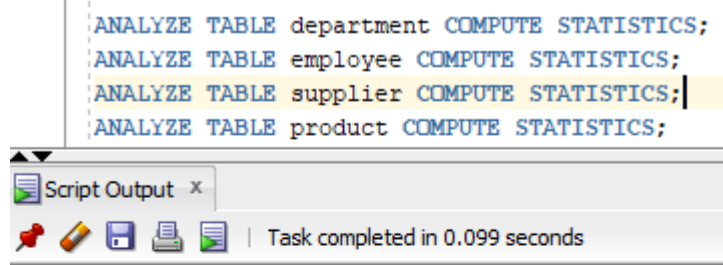


Table DEPARTMENT analyzed.

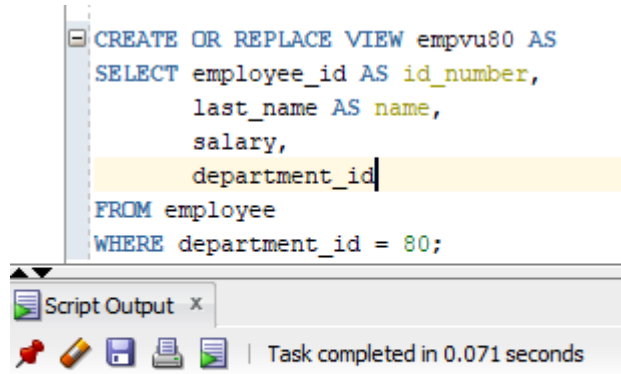
Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.

Table PRODUCT analyzed.

5.

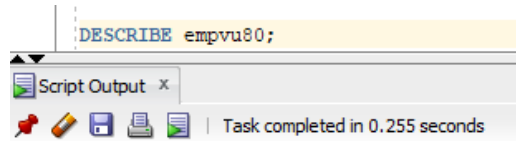
```
CREATE OR REPLACE VIEW empvu80 AS  
SELECT employee_id AS id_number,  
       last_name AS name,  
       salary,  
       department_id  
FROM employee  
WHERE department_id = 80;
```



View EMPVU80 created.

6.

```
DESCRIBE empvu80;
```



View EMPVU80 created.

Name	Null?	Type
-----	-----	-----
ID_NUMBER	NOT NULL	NUMBER
NAME	NOT NULL	VARCHAR2(50)
SALARY		NUMBER(10,2)
DEPARTMENT_ID		NUMBER

7.

```
DROP INDEX idx_emp_dept_id;  
CREATE INDEX idx_emp_dept_id ON employee(department_id);
```

Script Output x

     | Task completed in 0.053 seconds

Index IDX_EMP_DEPT_ID created.